



INTELLIGENT LOGISTICS ACCESS CONTROL FOR MARITIME PORTS

**EVERY JOURNEY STARTS WITH A
MICRO IDEA!**

PROBLEMATIC

Access to the maritime port suffers from congestion and lack of visibility. The APCS needs to regulate truck flows through a booking system that manages time slots and terminal capacities. However, managing these bookings requires constant interaction; carriers and operators need a simple way to query booking statuses, slot availability, or passage history without navigating complex interfaces.

SOLUTION

Participants must develop a Truck Booking Management System integrated with an Intelligent Logistics Assistant. The system must regulate port access by allowing carriers to book time slots while operators manage terminal capacities and validate requests. The AI agent, built on a multi-agent architecture, serves as a conversational Help Desk enabling users to check booking statuses, query slot availability, and retrieve historical data through natural language.



REQUIREMENTS

- **Multi-Role Logistics Portal:** Distinct Interfaces For Port Admins (Global Configuration), Terminal Operators (Capacity Management, Booking Validation), And Carriers (Booking Requests, Fleet Management), With Role-Based Access Control.
- **Capacity & Slot Engine:** Logic To Define Time Slots Per Terminal/Gate, Enforce Capacity Limits (E.G., Max N Trucks Per Hour), And Display Real-Time Availability To Prevent Overbooking.
- **Booking Lifecycle & QR:** A Complete Booking Flow Allowing Carriers To Create, Modify, And Cancel Bookings, With Clear Status Tracking (Pending / Confirmed / Rejected / Consumed) And Generation Of A Unique Booking Number Or QR Code Upon Validation.



REQUIREMENTS

- **Operational Dashboard & Notifications:** A Real-Time Monitoring View For Operators To Validate Incoming Requests And View Terminal Saturation, With Essential Notifications (Email Or In-App) For Booking Events, Delays, Or Capacity Alerts.
- **Logistics AI Agent:** An AI Assistant With Conversation History That Utilizes An "Orchestrator Agent" To Interpret Logistics-Specific Queries In Natural Language And Route Them To Specialized Sub- Agents (E.G., A "Booking Agent," A "Slot Availability Agent"), Answering Questions Like "Is There Space At Terminal A Tomorrow?"
- **Traceability & Logging:** Logging Of All Booking Modifications And AI Interactions To Ensure Auditability, Compliance With Data Protection Requirements.



REQUIREMENTS

- **Self-Service Status Checks:** The AI Must Return Structured, Contextual Answers (Booking Statuses, Slot Tables, Passage History) Respecting User Permissions, Enabling Carriers And Operators To Retrieve Data Without Navigating Menus.
- **Future Integration:** And A Base REST API Prepared For Future Integration With The GateControl System



TECHNICAL SUGGESTIONS

- **Frontend:** Modern JavaScript Framework (Example: Angular, React)
- **Backend:** Robust, Modular Backend Framework (Example: Spring Boot, Laravel)
- **Database:** Relational Database Management System (PostgreSQL, MySQL, Or Similar)
- **Deployment:** Containerization And DevOps Readiness (Docker, CI/CD Pipeline, Kubernetes Orchestration)

Note: The Mentioned Technologies Serve As Illustrative Examples And Are Not Mandatory. Nevertheless, The Project Must Be Fully Containerized (Docker) And Accompanied By Comprehensive Documentation.



JUDGING CRITERIA

Judge Type	Category	Criteria	Points	Total
Main	Innovation & Problem-Solving	Originality Of Solution, Feature Innovation, Problem Understanding	15	40
	UI/UX & User Experience	Interface Design Quality, Usability & Intuitiveness, User Experience Flow	10	
	Real-World Applicability	Practical Value For Port Stakeholders, Scalability & Multi-Organization Support, Deployment Readiness	10	
	Presentation & Pitching	Pitch Clarity & Impact, Demonstration Quality	5	
Technical	Functionality & Completeness	Core Requirements Coverage, Feature Robustness, Advanced Features Implementation	25	60
	Technical Excellence	Code Quality & Architecture, Technology Stack Choice & Implementation, Scalability & Performance	20	
	Security & Compliance	Data Protection, Audit & Traceability	15	
				Total : 100